# REAPER-Notator v1.9.2

Comprehensive Codebase Volume & Structural Analysis

| TOTAL LINES | EXECUTABLE CODE | SOURCE FILES | PATTERN LIBRARY |
|---|---|---|---|
| 56,202 | 47,325 | 75 Lua | 1,200 JSON |

## 1. Architecture & Module Volume Breakdown

| Module Directory | Files | Total Lines | Code Lines | Comments | Blank | Code Ratio |
|---|---|---|---|---|---|---|
| modules/services | 28 | 29,218 | 24,931 | 1,788 | 2,630 | 85.3% |
| modules/rendering | 7 | 11,709 | 9,956 | 723 | 1,082 | 85.0% |
| modules/ui | 19 | 9,337 | 7,539 | 718 | 1,106 | 80.7% |
| modules | 3 | 2,186 | 1,908 | 417 | 138 | 87.3% |
| modules/input | 2 | 2,002 | 1,740 | 91 | 173 | 86.9% |
| Root | 1 | 976 | 654 | 248 | 78 | 67.0% |
| modules/classes | 15 | 774 | 597 | 110 | 96 | 77.1% |
| **TOTAL** | **75** | **56,202** | **47,325** | **4,095** | **5,303** | **84.2%** |

## 2. Key Core System Files by Volume (Top 15)

| File Name | Subsystem / Module | Lines | Code | Comments |
|---|---|---|---|---|
| midi_service.lua | modules/services | 5,994 | 5,226 | 241 |
| score_canvas.lua | modules/rendering | 4,434 | 3,844 | 210 |
| engraver.lua | modules/rendering | 2,789 | 2,375 | 213 |
| canvas_context_menus.lua | modules/rendering | 2,204 | 1,913 | 105 |
| slur_service.lua | modules/services | 2,120 | 1,830 | 102 |
| musicxml_import_service.lua | modules/services | 2,118 | 1,839 | 124 |
| musicxml_export_service.lua | modules/services | 1,698 | 1,435 | 145 |
| portamento_service.lua | modules/services | 1,549 | 1,312 | 88 |
| glissando_service.lua | modules/services | 1,474 | 1,245 | 79 |
| midi_editor_drawer.lua | modules/ui | 1,448 | 1,205 | 118 |
| settings_modal.lua | modules/ui | 1,409 | 1,144 | 96 |
| mouse_handler.lua | modules/input | 1,319 | 1,157 | 50 |
| dynamics_engine.lua | modules/services | 1,275 | 1,128 | 45 |
| state.lua | modules | 1,132 | 980 | 199 |
| clipboard_service.lua | modules/services | 1,044 | 890 | 58 |

## 3. Curated Orchestral Pattern Library (1,200 Patterns)

| Category Folder | Genre / Instrumentation Scope | Pattern Count |
|---|---|---|
| 01_Strings_Staccato | Cinematic Strings (Staccato / Spiccato Ostinatos) | 150 patterns |
| 02_Strings_Pizzicato | Orchestral Strings (Pizzicato / Bartók Snap) | 150 patterns |
| 03_Brass_Blockbuster | Epic Brass (Horns, Trumpets, Trombones, Tubas) | 150 patterns |
| 04_Cinematic_Melodies | Heroic Themes & Symphonic Lead Melodies | 150 patterns |
| 05_Counter_Melodies | Polyphonic Counter-Melodies & Descant Voices | 150 patterns |
| 06_Woodwinds_Textures | Woodwind Runs, Flutter-Tongue & Textures | 150 patterns |
| 07_Cinematic_Piano | Concert Grand Piano (Arpeggios & Ostinatos) | 150 patterns |
| 08_Ancient_Harp_Greek_Roman | Antiquity Concert Harp (Greek & Roman Hymns) | 150 patterns |
| **TOTAL PATTERNS** | **8 Curated Production Categories** | **1,200 patterns** |

## 4. Specialized Orchestral & Engraving Services Subsystem

| Service Subsystem File | Musical & Architectural Responsibility | Code Lines |
|---|---|---|
| divisio_vocum_service.lua | Divisio Vocum - Orchestral Voice Division across tra | 308 |
| dynamics_engine.lua | Orchestral Dynamics Engine - Bezier/Linear CC automa | 1,128 |
| glissando_service.lua | Glissando Engine - Smooth chromatic / diatonic slide | 1,245 |
| midi_service.lua | Core MIDI Engine - Low-latency take I/O and event sy | 5,226 |
| musicxml_export_service.lua | MusicXML 4.0 Exporter - Standard score interchange f | 1,435 |
| musicxml_import_service.lua | MusicXML 4.0 Importer - Orchestral part loading and | 1,839 |
| audio_preview.lua | Audition Preview Engine - Non-destructive note click | 805 |
| reaticulate_parser.lua | Reaticulate Bridge - Bank parsing and articulation k | 376 |
| scale_service.lua | Scale & Harmony Service - Chord Track harmonic trans | 835 |
| slur_service.lua | Slur Engine - Ergonomic Bezier phrasing curves and k | 1,830 |
| portamento_service.lua | Portamento Service - High-resolution micro-tonal pit | 1,312 |
| pattern_service.lua | Pattern Engine - Orchestral motif browser and timeli | 411 |
| rehearsal_mark_service.lua | Rehearsal Marks - Structural navigation and drag-and | 244 |
| fermata_service.lua | Fermata Service - Score-wide pause holds and tempo m | 232 |
| pedal_service.lua | Pedal Service - Classical damper and sustain pedal C | 530 |

**Quality Assurance & Standards Status: 100% Certified**

All 75 source files and 1,200 pattern JSON files compiled with 0 errors. Fully compliant with Gardner Read & Elaine Gould engraving standards.